Acta Neurologica Belgica
https://doi.org/10.1007/s13760-023-02466-w

REVIEW ARTICLE

# A review of the personality traits in post-concussion syndrome

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Received: 15 May 2023 / Accepted: 18 December 2023
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## Abstract

**Introduction** Various manifestations ranging from physical symptoms to cognitive and emotional impairments could often be seen following head concussions that lead to mild traumatic brain injury (mTBI). These symptoms are commonly comprising the post-concussion syndrome (PCS) and their resolution could be influenced by multiple factors. Personality traits have been suggested as potential risk factors for the emergence and persistence of PCS. In this study, we aimed to investigate the possible predisposition to PCS given by certain personality traits.

**Methods** Prospective cohort studies, observational studies, and cross-phenotype polygenic risk score analyses were selected from the main scientific databases (PubMed/Medline, Scopus, EMBASE, Web of Science) based on multiple-step screening, using keywords (such as "personality traits", "post-concussion syndrome", "traumatic brain injury", "anxiety", "depression", "resilience", and "somatization") and inclusion/exclusion criteria (English written studies available in full text presenting relevant data on TBI patients and their personality traits; reviews, animal studies, and studies not written in English, not available in full text, or not presenting full demographical and clinical data were excluded). The investigated personality traits included emotional reserve, somatic trait anxiety, embitterment, mistrust, parental anxiety, state anxiety, trait anxiety, anxiety sensitivity, pain catastrophizing, helplessness, sports-concussion symptom load, and cognitive resilience.

**Results** The reviewed data from 16 selected studies suggested that personality traits play an essential role in the development and persistence of PCS. Emotional reserve, cognitive resilience, and lower levels of somatic trait anxiety were associated with better outcomes in PCS. However, higher levels of anxiety sensitivity, pain catastrophizing, helplessness, and sports-concussion symptom load were associated with worse outcomes in PCS. Parental anxiety was not associated with persistent symptoms in children following concussion. Despite the statistical analysis regarding the included publications bias was low, further studies should further investigate the correlation between TBI and some personality traits, as some of the selected studies did not included healthy individuals and their psychological profiles for comparison and correlation analysis.

**Conclusion** Personality traits may help predict the development and persistence of PCS following mTBI. Understanding the personality traits roles in PCS could assist the development of targeted interventions for the prevention and treatment of PCS. Further research is needed to better understand the complex interactions between personality traits, neurobiological factors, and psychosocial factors in PCS.

**Keywords** Post-concussion syndrome · Mild traumatic brain injury · Personality traits · Anxiety · Emotional reserve · Cognitive resilience

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Published online: 09 January 2024

## Introduction

Mild traumatic brain injury (mTBI), also known as concussion, is a significant public health concern due to its high incidence and potential long-term consequences, including the development of persistent post-concussion syndrome (PCS) [1]. Despite that statistical reports are available for just a few geographic areas of the globe, Dewan et al. [2] recently estimated that 64 to 74 million new cases of all-causes TBI are occurring annually, with more than 80% being mTBIs. A recent systematic review that included studies with 101 to 208,195 patients reported a mean value for TBI incidence of 258 per 100,000 population per year, and mortality rates up to 24.4 per 100,000 population per year, as analysed from regional-level studies [3]. Furthermore, recent estimates suggested that more than 40% (but less than 80%) of the TBI patients will experience PCS symptoms [4].

TBI is currently defined as a form of acquired brain injury due single unpredicted trauma (blow, bump, jolt, penetration) resulting in brain damage [5–7].

PCS is characterized by a range of symptoms, including headaches, dizziness, fatigue, cognitive deficits and mood disturbances that persist beyond the expected recovery period [8]. While various factors contribute to the development and persistence of PCS, recent research has highlighted the potential role of personality traits in predicting and influencing the outcome of PCS [9].

Personality traits refer to consistent patterns of thoughts, feelings, and behaviours that condition the response of an individual to stressors and environmental challenges [10]. They can also refer to any aspect of the personality of an individual, including beliefs, values, attitudes, and other aspects that may be more situation-specific or transient [11]. In this context, personality traits are responsible of providing emotional backgrounds for the behavioural response when different environmental stimuli are applied [12]. Thus, personality traits refer to the general envisage of one's behaviour or state of mind and to a behavioural response that occurs in certain situations and could differ from one stimulus to another. To provide efficient prevention and care interventions that would improve the outcomes for individuals experiencing persistent symptoms after mTBI, it could be important to identify the specific personality traits that are associated with PCS. Thus, this study aimed to review the current literature on personality traits in PCS, highlighting the potential mechanisms underlying their association and their implications for clinical practice.

## Materials and methods

### Literature search

A comprehensive literature search was conducted on PubMed, Embase, and PsycINFO databases up to April 2023. The following search terms were used: "personality traits", "post-concussion syndrome", "traumatic brain injury", "anxiety", "depression", "resilience", and "somatization" in combination with Boolean operators OR and AND for PubMed and PsychINFO databases searches (("traumatic brain injury" OR "post-concussion syndrome") AND ("personality traits" OR "anxiety" OR "depression" OR "resilience" OR "somatization")), and multiple-line search for Embase database (1 exp traumatic brain injur*.mp. 2 exp "post-concussion syndrome".mp. 3 or/1–2 4 exp "personality trait*"/5 exp "anxiety"/ 6 exp "depression" 7 exp "resilience" 8 exp "somatization" 9 or/4–8 10 (human not (animals).sh. 11 3 and 9). The search was limited to studies published in English and focussed on human patients. Studies were included if they investigated the relationship between personality traits and PCS or persistent PCS. The reference lists of selected studies were also reviewed for additional relevant studies.

### Study selection

Two authors independently reviewed the titles and abstracts of the identified studies and reviewed the full text to determine their eligibility for inclusion in this review. Inclusion criteria were considered: (1) studies involving human patients (both adults and adolescents/children) with TBI or PCS (disregarding the TBI cause); (2) presenting relevant information on the psychological status and personality traits at the time of traumatic event, and in follow-up (if available, up to 1 year); (3) available in full text; (4) written in English. Exclusion criteria: (1) reviews or animal models studies; (2) not presenting psychological information; (3) not presenting demographical data (including socio-cultural context).

### Data extraction

Data were extracted from the selected studies, including study design, sample size, participant characteristics, assessed personality traits (anxiety, depression, resilience, somatization, embitterment, mistrust, alexithymia, pain catastrophizing, helplessness, intrapersonal emotional reserve, neuroticism), outcome measures (correlations between occurrence of TBI/ PCS, symptoms duration or severity, symptoms persistence, and symptoms additions versus personality traits), and main findings. We generally followed the

PICO diagram (patient Group: patients with mTBI followed by PCS; intervention: personality traits; comparison: mTBI patients not developing PCS; outcome: increased occurrence of some personality traits in those with PCS).

## Data synthesis

Due to the heterogeneity of the studies in terms of design, outcome measures, and assessed personality traits, a narrative synthesis was conducted. The studies were grouped according to the assessed personality traits, and the findings were summarized.

## Quality assessment

The quality of the selected studies was assessed using the Cochrane Risk of Bias tool for randomized controlled trials and the Newcastle–Ottawa Scale for non-randomized studies. Two authors independently assessed the quality of the studies. Discrepancies in opinion, if any, found their resolution through mutual consensus.

## Results

Once the initial search was completed, duplicate articles were removed, and the remaining articles were screened by two authors. Articles that did not meet the selection criteria, such as reviews or case reports, were removed. After this screening process, a total of 20 studies were re-screened. Finally, the authors selected 16 studies that met the final inclusion criteria for the review (Fig. 1). There were no concerns on the quality assessment of the studies (Fig. 2). A summarization of the selected studies is available in Table 1.

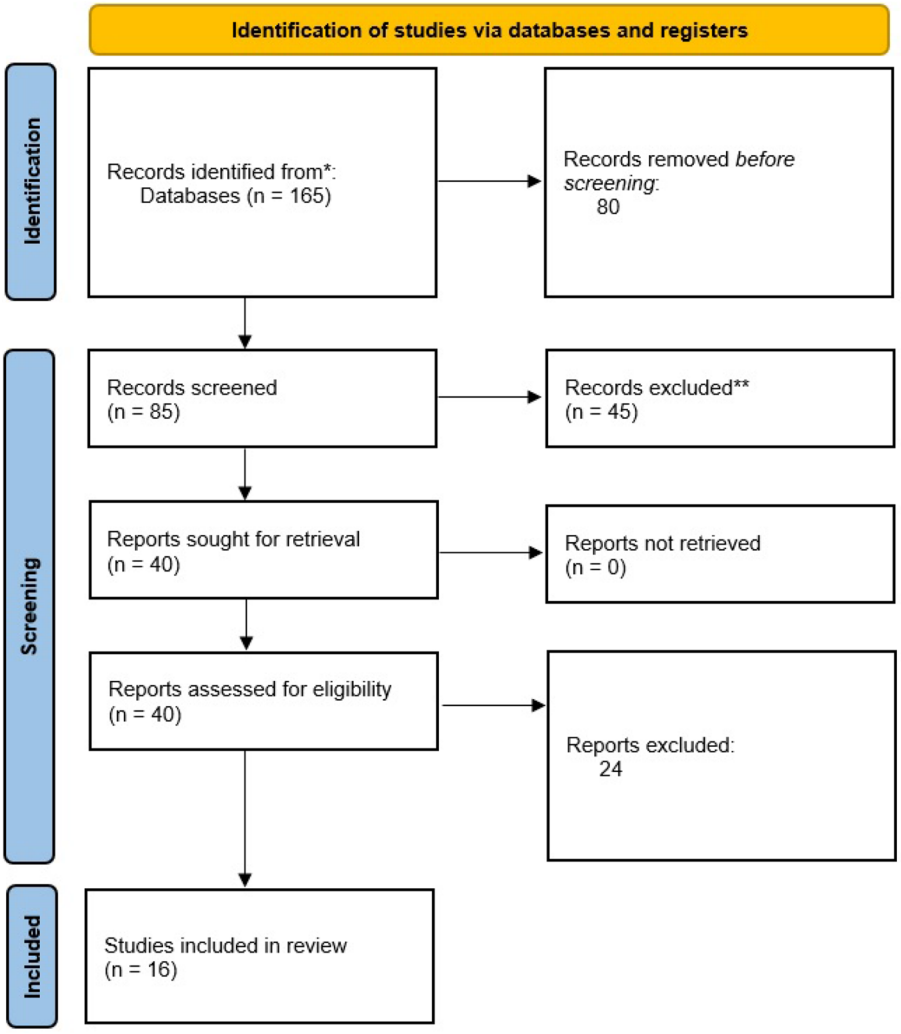


**Fig. 1** Studies selection flowchart (* - initial screening of the databases based on keywords; ** - applied criteria for exclusion: studies not written in English, not available in full text, not focussing on human individuals, and not presenting patients demographical or psychological traits)

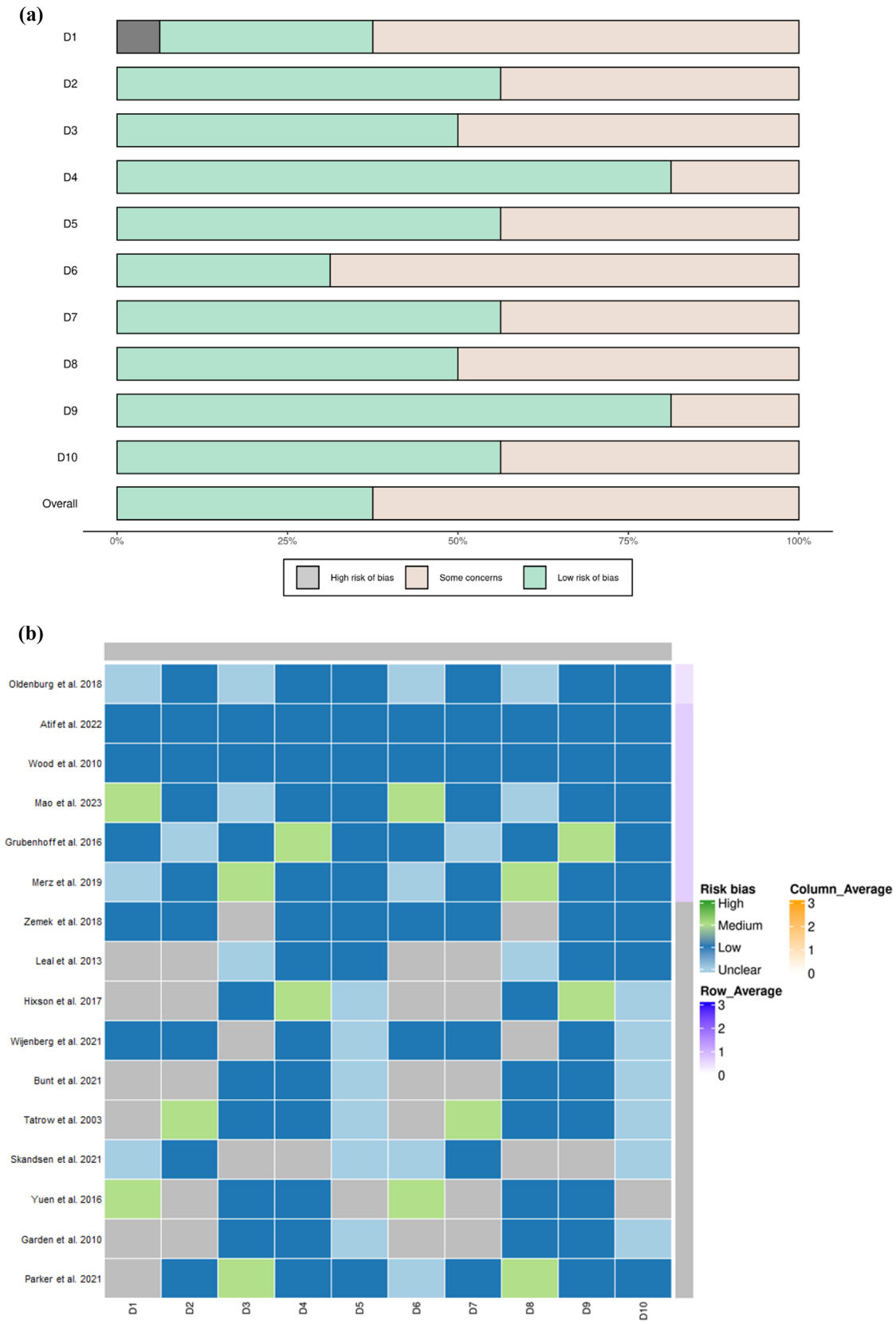


**Fig. 2** Quality assessment of the selected studies

**Table 1** Summarization of findings in the selected studies

| Study | Participants | Follow-up period | Personality evaluation | Results | Associated personality trait |
|---|---|---|---|---|---|
| [9] | 42 with PPCS (>4 weeks), 55 healthy adolescents (aged 15 ± 2 years) | | 10-item big five inventory and the child<br>Adolescent social support scale | Higher neuroticism scores in persistent PCS group;<br>Less social support in persistent PCS group | Neuroticism |
| [13] | 122 patients with mTBI (blunt head trauma)<br>(Aged 15–65 years) | 1 week after the injury<br>1 year after the injury | The Swedish universities scales of personality<br>The sense of coherence scale<br>Alcohol use disorders identification test | Acute stage of mTBI in patients later developing PCS included symptoms such as anxiety and depression ($p < 0.001$, as compared to those who recovered), as well as previous or concurrent psychological disorders ($p < 0.001$);<br>1-year follow-up: 12% of the participants were diagnosed with persistent PCS, also exhibiting increased levels of stress ($p = 0.018$) | Somatic trait anxiety<br>Embitterment<br>Mistrust<br>Low psychological resilience |
| [13] | 98 children (sports-concussion) (aged 5 to 17 years) and their parents | 3-day, 7-day, 2-week, 1-month, and 3-month follow-ups | Spielberger State-Trait Anxiety Inventory-state anxiety scale | At 1 month, parental anxiety was not associated with PCS ($p = 0.63$), but with previous diagnosis of ADHD ($p = 0.001$);<br>Parental anxiety was significantly higher in persistent PCS children ($p = 0.001$) | Parental anxiety |
| [15] | 61 mTBI patients (falls and assaults)<br>61 Controls | 2 weeks post-TBI | 20-item Toronto alexithymia scale,<br>Anxiety sensitivity index,<br>State-Trait Anxiety Inventory | mTBI was associated with anxiety state ($p = 0.005$), alexithymia ($p = 0.005$), and psychological distress ($p = 0.005$), as compared to controls;<br>Post-concussion symptoms were associated with anxiety state and alexithymia, as well as with increased psychological distress ($p = 0.005$) | State anxiety<br>Trait anxiety<br>Depression<br>alexithymia |
| [16] | 37 adults with mTBI (falls, motor vehicle accidents, sports-related injury) and acute post-traumatic headache | 6 months post-TBI | Beck depression inventory<br>State-Trait Anxiety Inventory<br>Pain catastrophizing scale<br>Primary care post-traumatic stress disorder screen | From the top 5 most predictive variables for headache improvement/non-improvement at 3 months included were helplessness, and trait anxiety | Catastrophizing<br>Trait anxiety |
| [17] | 80 healthy adults | | Post-concussion catastrophizing scale<br>Fear of mental activities scale<br>State-Trait Anxiety Inventory<br>Beck depression inventory-II revised | Association between PCS-like symptoms and catastrophizing ($r = 0.31$, $p < 0.01$, Spearman correlation) | Catastrophizing |

**Table 1** (continued)

| Study | Participants | Follow-up period | Personality evaluation | Results | Associated personality trait |
|---|---|---|---|---|---|
| [18] | 35 mTBI patients (motor vehicle accidents)<br>32 Headache patients | | Minnesota multiphasic personality inventory,<br>Daily stress inventory,<br>Daily hassles scale,<br>Beck depression inventory,<br>State-trait anxiety inventory,<br>State-Trait Anger Expression Inventory | Significant differences between mTBI patients with post-traumatic headache and control group: state and trait anxiety, beck depression inventory ($p<0.01$)<br>Significant differences between mTBI patients with post-traumatic headache and mTBI patients without post-traumatic headache: hypochondriasis ($p<0.01$), hysteria ($p<0.01$), mania ($p<0.01$), and social introversion ($p<0.01$) | State and trait anxiety<br>Depression |
| [19] | 179 children with concussion (Age 8–18 years) | 1 month | State-Trait Anxiety Inventory for Children<br>Personality inventory for children-2 | State-trait anxiety inventory ($p=0.04$) and somatization subscale ($p=0.01$) with worse results in children with delayed resolution of symptoms | State-trait anxiety<br>Somatization |
| [20] | 110 concussed individuals<br>502 Without any history of concussions (aged between 18 and 89 years) | | IPIP-NEO-120 questionnaire | Significant difference between positive and negative concussion was obtained for neuroticism facet scores ($p=0.018$)<br>Significantly decreased agreeableness was also reported in negative history group, as compared to positive history of concussions group ($p=0.018$) | Neuroticism<br>Agreeableness |
| [21] | 378 mTBI patients,<br>82 Trauma controls,<br>81 Community controls | 3 months post-TBI | Big five inventory<br>Life orientation test-revised<br>Resilience scale for adults | Resilience and neuroticism were reported as a strong predictor of PCS ($p<0.01$) | Resilience<br>Neuroticism |
| [22] | 332 mTBI patients (sports-related concussions)<br>Adolescents and young adults (aged 13 to 25 years) | 3 months post-TBI | Brief resilience survey | Initial visit: lower resilience ratings, increased anxiety and depression;<br>3-month follow-up: lower resilience determined delayed recovery, aggravated symptoms | Resilience<br>Anxiety<br>Depression |
| [23] | 53 mTBI patients<br>53 Healthy participants | 4 months | Checklist of post-concussion symptoms and the health, personality, and habit scale | Pre-injury personality traits (anxiety and depression) significantly positive association with PCS | Anxious and depressive-like personalities |
| [24] | 96 healthy participants (adults, > 18 years) | | Millon clinical multiaxial inventory III scale | Negative affect, depression, dysthymia, anxiety, sadistic, somatic, and borderline tendencies were correlated with PCS symptoms | Negative affect<br>Depression<br>Anxiety<br>Somatization<br>Sadistic and borderline tendencies |

**Table 1** (continued)

| Study | Participants | Follow-up period | Personality evaluation | Results | Associated personality trait |
|---|---|---|---|---|---|
| [25] | 75 mTBI patients<br>79 orthopedic trauma controls<br>(adults, > 18 years) | 3 months | Multidimensional personality questionnaire<br>11-Item disinhibition scale was used<br>Personality disorder inventory for DSM-5–brief form | Anxiety ($p < 0.05$) and somatization ($p < 0.05$) were strong predictors for PCS (at 3-months follow-up);<br>Negative affect/emotionality ($p < 0.05$), psychoticism ($p < 0.01$), and detachment ($p < 0.05$) predicted chronic mTBI symptoms | Anxiety<br>Somatization<br>Psychoticism<br>Detachment<br>Negative affect |
| [27] | 75 male volunteers<br>(Young adults, 18 to 30 years) | 2 days | State-Trait Anxiety Inventory<br>Self-evaluation of tension level<br>Social phobia inventory<br>Hospital anxiety and depression scale<br>Dental anxiety scale | Trait anxiety modulates anxious response (state anxiety);<br>Trait anxiety modulates reactive anxiety;<br>Trait anxiety correlates with state anxiety in social threat premises ($r = 0.74$) | Trait anxiety<br>Social anxiety |
| [28] | 329 athletes with mTBI from 3 studies<br>(Adults, > 18 years) | 7 days to up to 1 year | Modified 47-item Holmes, and Rahe social readjustment Rating scale<br>State-Trait Anxiety Inventory<br>20-Item Toronto alexithymia scale<br>Anxiety sensitivity index<br>Beck depression inventory-II | mTBI patients reporting PCS symptoms were positively assessed with anxiety sensitivity and alexithymia, as compared to controls ($p = 0.005$);<br>mTBI patients with state anxiety also reported depression (95% CI);<br>Trait and state anxieties were not differently exhibited between the groups ($p = 0.193$) | Anxiety Sensitivity<br>Alexithymia<br>State anxiety<br>Depression |

## Discussion

### State anxiety, trait anxiety, and anxiety sensitivity & alexithymia

State anxiety refers to the temporary feeling of anxiety or worry that is experienced in response to a particular situation or event. Trait anxiety refers to a person's overall tendency to experience anxiety across different situations or events. Anxiety sensitivity refers to the degree to which a person fears the physical symptoms of anxiety such as heart palpitations, sweating, or shortness of breath [27]. People with high anxiety sensitivity tend to interpret these bodily sensations as dangerous or harmful, leading to increased anxiety and avoidance of situations that might trigger such sensations.

Several studies have investigated the role of different anxiety measures in PCS. In a meta-analysis, Hixson et al. [28] reported that state and trait anxiety were unrelated to the increased post-concussion symptom scores. Still, greater anxiety sensitivity was related to higher post-concussion symptom scores, which may manifest as somatic symptoms following concussion. They also revealed that anxiety sensitivity might be a risk factor for symptom development.

On the other hand, somatic trait anxiety refers to a tendency to experience anxiety in response to physical sensations or symptoms, such as headaches or dizziness, which are common after mTBI (prevalences: headache, 47–95% of mTBI patients after less than 3 months, 14–58% of mTBI patients after 1 year [29, 30]; dizziness, up to 26% of the TBI patients at 1 year, and 50% of TBI patients at up to 5 years [31, 32]). In this context, Oldenburg et al. found that somatic trait anxiety was higher in patients with persistent PCS, as compared to those who recovered ($p < 0.001$) [13] suggesting that somatic trait anxiety may play a role in the emergence and persistence of PCS after mTBI.

Regarding anxiety sensitivity and alexithymia implications in the development of psychological distress and post-concussion symptoms after mTBI, Wood et al. [15] evaluated 61 patients who suffered from mTBI and found that anxiety sensitivity, alexithymia, and psychological distress scores were significantly correlated with PCS scores, as compared to demographically matched controls ($p = 0.005$). Also, the increased anxiety sensitivity, as well as of alexithymia, were associated with a greater number of post-concussion symptoms and increased psychological distress. In addition, Wood et al. [8] also explained the possible implication of the combination between low mood and increased anxiety sensitivity, as observed by over half of the variance in post-concussion symptom reporting in the mTBI group, while the combination of trait anxiety, alexithymia, and post-concussion symptoms explained over three-quarters of the variance in levels of mood. Thus, it seems that anxiety sensitivity and alexithymia could play a mediating role in the development of post-concussion symptoms and psychological distress after mTBI and suggested that the early identification of individuals with high levels of anxiety sensitivity and alexithymia could be an important prevention intervention against PCS occurring after mTBI.

Another study focussing on post-traumatic headache (PTH) [18] found that the patients that experienced PTH exhibited significantly poorer functioning and outcomes (as evaluated by Psychosomatic Symptom Checklist, Post-concussion Syndrome Checklist, Axis II Psychiatric Diagnoses, Minnesota Multiphasic Personality Inventory, and the Daily Hassles Scale), as compared to multiple control groups ($p < 0.01$ for state and trait anxiety, depressive symptoms, and manic manifestations). In addition, the higher scores the PTH patients obtained for Axis I Psychiatric Diagnoses, Beck depression inventory, State-Trait Anxiety Inventory, and State-Trait Anger Expression Inventory suggested that the presence of specific cognitive and affective distress raising the awareness for the need of better assessment and treatment of PTH in mTBI, as well as the possible predisposition to prolonged PCS and complications in some affective impairments context.

### Parental anxiety

Zemek et al. [14] investigated the association between parental anxiety and persistent PCS in children following a concussion and found that despite that parental anxiety in acute paediatric concussion was not associated with children's persistent symptoms at 1 month, the parents of persistently symptomatic children remained significantly more anxious than those whose children's symptoms had resolved. This observation could suggest that although parental anxiety may not be a predictor of persistent PCS in children, it may have an impact on the mental health and well-being of the parents themselves.

### Embitterment, mistrust, and neuroticism

Embitterment refers to a persistent negative emotional reaction to a perceived injustice or failure, and mistrust refers to a general lack of trust in others, and they are both components of neuroticism manifestations. These personality traits were identified as intrapersonal emotional factors that shape the emergence and persistence of PCS after mTBI, as also shown by Oldenburg et al. [13] (higher levels of embitterment and mistrust in PCS patients, as compared to those who

recovered, $p < 0.001$). Also, somatic trait anxiety and anxiety sensitivity could be construct related to neuroticism [13, 28].

Neuroticism is a personality trait that is consistently linked to an increased risk of developing psychological distress and anxiety-related disorders. Atif et al. [9] reported that adolescents with persistent PCS receive higher neuroticism scores (Big five inventory), as compared to healthy peers. This could suggest that they may be at a greater risk of experiencing persistent symptoms following a concussion and thus neuroticism could be a useful target during PCS management and treatment.

Furthermore, Merz et al. [20] reported significant relationships between concussion symptoms occurrence and personality traits of neuroticism, and other personality components ($p = 0.018$, positive versus negative history of concussion). However, Skandsen et al. [21] suggested that the differences residing in psychiatric distress (such as depressive symptoms) accounted for a significant degree of variance in symptom reporting and could likely be strong influencers in recovery trajectory. Thus, the role of neuroticism in PCS could be rather complex and depend on many factors, especially on symptoms intensity or duration that are often based on self-reporting.

## Pain catastrophizing, helplessness, and state-trait anxiety

Pain catastrophizing is the tendency to exaggerate the negative aspects of pain and to feel helpless and overwhelmed by it [33]. Helplessness is a state of mind in which a person feels powerless to influence or change their situation [34] and it is often associated with feelings of depression and anxiety, and poor problem-solving and coping behaviours.

Pain catastrophizing, helplessness, state-trait anxiety, as well as sports-concussion symptom load were all previously associated with PCS, as Mao et al. [16] found. Specifically, the Pain Catastrophizing Scale total score and helplessness sub-domain score, Sports Concussion Assessment Tool Symptom Evaluation total score and number of symptoms, and State-Trait Anxiety Inventory score were among the top five contributing features for predicting headache improvement and trajectory over the first 3 months after mTBI, and could efficiently help predict the recovery from PTH. Furthermore, using a randomized crossover within-subject design with 80 participants exposed to demanding cognitive tasks and their simplified counterparts, Wijenberg et al. [17] investigated whether catastrophizing about mental activities are related to fear-avoidance behaviour in healthy adults and concluded that fear and avoidance of certain activities can lead to the development of symptoms similar to those seen in patients with mTBI. The results showed significant correlations between fear-avoidance model and catastrophizing by relation to behavioural avoidance of cognitively challenging tasks in the community sample ($p < 0.01$).

## Intrapersonal emotional reserve

Increased intrapersonal emotional reserve could lead to better emotional equipment in handling stress and difficult situations and the predisposition to positive outlook and emotional balance along the course of aversive emotional stimulation and could play an important role in recovery from mTBI [13]. Oldenburg et a [13] described the correlations between PCS and lower levels of psychological resilience ($p < 0.018$, at 1-year follow-up, mTBI patients that recovered versus mTBI patients exhibiting PCS symptoms). In other words, those who had better emotional reserve before the injury were less likely to develop and experience persistent PCS after mTBI. Also they suggested that psychological resilience could interact with personality traits such as somatic trait anxiety, embitterment, and mistrust in influencing the recovery process.

Otherwise, psychological resilience—another component of emotional reserve—was found to be a contributing factor in predicting PCS in patients with mTBI and of significant importance in planning interventions for these patients [21]. Thus, Skandsen et al. [21] recently added resilience in the predictive model for PCS along with other personal factors, such as working less than full time before injury, pre-injury pain, poor sleep quality, and being female as they observed that patients with lower levels of resilience were more likely to report PCS. Oldenburg et al. [4] also reported a possible association between multiple personality traits (somatic trait anxiety, embitterment, mistrust, and psychological resilience) as implicated in the delay of PCS symptoms resolution [4].

Also, psychological resilience was shown to be important in adolescents and young adults (13–25 years) affected by sport-related concussions by relation to symptoms resolution [22]. While initial evaluation correlated lower resilience ratings (as measured by Brief resilience survey) with greater number and severity of PCS symptoms, during the 3-month follow-up, it was showed that the patients with lower resilience had greater aggravation of symptoms when engaged into physical of cognitive activities. However, the study limitations hindered further analysis on the extent to which resilience measured after sport-related concussions reflects pre-injury characteristics and the possible use of psychological resilience enhancement to promote recovery.

## Somatization

Somatization is a complex phenomenon referring to the psychological distress or emotional problems that are exhibited as physical symptoms, such as pain, fatigue, or digestive

issues, without any identifiable medical cause [35]. Trauma, stress, anxiety, psychiatric disorders, such as somatic symptom disorder or conversion disorder could lead to somatization manifestations.

Regarding the implication of somatization in mTBI and PCS, Hixson et al. [28] explained that greater anxiety sensitivity experienced by the mTBI patients could lead to somatic symptoms. Oldenburg et al. [13] also reported physical symptoms of anxiety in patients with PCS showing higher somatic trait anxiety, as compared to those who recovered from mTBI ($p < 0.001$). These symptoms included tension headaches, muscle pain, dizziness, and fatigue reported as frequent in PCS and in those manifesting psychological anxiety and depression.

On the other hand, somatization could be associated with delayed symptom resolution, as shown in a study focussing on children with concussions [19]. Grubenhoff et al. [19] found that children with delayed symptom resolution had worse scores on the State-Trait Anxiety Inventory for Children and the somatization subscale, as compared to those with early symptom resolution ($p = 0.04$, and $p = 0.01$, respectively) suggesting that children with PCS symptoms persisting for at least 1 month after concussion could benefit from neuropsychological care alongside typical PCS management [19].

## Other personality traits

Yuen et al. [23] found a significant positive association between PCS and pre-injury personality traits, specifically those related to depression and anxiety. They reported that the patients that manifested depressive or anxious-like personality developed more PCS symptoms, as compared to those with different personality traits. This could suggest that some pre-existing personality traits contribute to increasing the risk of developing PCS after mTBI. In this context, it could be relevant to assess the personality traits in patients with mTBI when attending to understand and manage the development of PCS.

A recent study involving 96 healthy participants aimed to examine the relationship between personality characteristics (as evaluated by Millon Clinical Multiaxial Inventory III) and self-reported post-concussion-like symptoms (as evaluated by post-concussion symptom inventory) [24]. Besides the fact that about half of the sample met the International Classification of Diseases-10 Criterion C symptoms for PCS, these patients received significantly higher scores on several MCMI-III items (negativistic, depression, major depression, dysthymia, anxiety, dependent, sadistic, somatic, and borderline scales). In this context, personality assessment could be useful in identifying patients who are at risk for PCS since some personality characteristics were seen to significantly contribute to PCS development (as evaluated by self-reported post-concussion symptoms inventories) [24].

Furthermore, Parker et al. even investigated the extent to which personality traits and psychiatric symptom severity could predict mTBI recovery [25]. Mediation modelling was performed on chronic symptoms evaluation at 3 months after mTBI using the typical Sport Concussion Assessment Tool in 75 patients with mTBI and 79 controls (diagnosed with orthopaedic trauma) and revealed that trait psychoticism directly predicted chronic mTBI symptoms ($p < 0.01$), making it the strongest personality predictor. In addition, negative affect/emotionality and detachment predicted chronic mTBI symptoms by indirectly enhancing the acute somatic complaints, mechanism also seen in orthopaedic trauma patients.

As our study aimed to review the role of personality traits in persistent PCS, we found that the recent reports mention that several personality traits could be associated with the emergence and persistence of PCS. Specifically, individuals with lower psychological resilience, higher somatic trait anxiety, embitterment, and mistrust are more likely to develop PCS following mTBI and recover slower (Fig. 3). However, the studies on parental anxiety suggested that the risks posed by personality could only be attributed to the patients traits despite that some cognitive patterns are acquired through mimicking and learning during childhood [14, 26]. Still, the persistent PCS developed in their children could significantly contribute to increased anxious behaviours and negatively impact the wellbeing of the parents.

The previous studies suggested that state and trait anxiety are not significant risk factors for the development of PCS. By contrast, anxiety sensitivity may contribute to

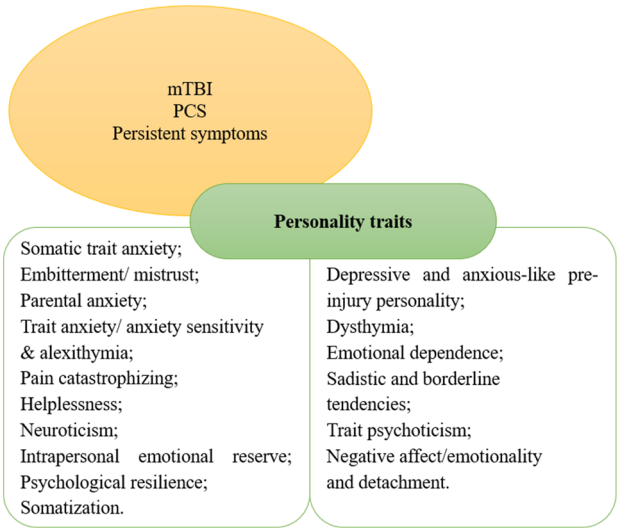


**Fig. 3** Personality traits in post-concussion syndrome (mTBI mild traumatic brain injury, PCS post-concussion syndrome

the development of PCS, particularly to the occurrence of somatic symptoms. Furthermore, pain catastrophizing and helplessness, sports-concussion symptom load, and state-trait anxiety were found as potent predictive factors of headache improvement in individuals with acute headaches due to mTBI.

### Limitations and future perspectives

While the studies discussed in this review provide valuable insight into the potential role of personality traits in PCS risk, there are several limitations to consider. For an instance, the small sample sizes could contribute to limit the generalizability of the findings. In addition, most studies rely on self-report measures which could be the subject to bias or inaccuracy. In this way, future research should focus on using larger and more diverse samples, as well as objective measures of personality traits, to better understand the role of these factors in PCS risk. In addition, further investigation into the underlying mechanisms linking personality traits and PCS risk, such as the influence of psychological distress, is needed.

## Conclusion

Personality traits could be involved in the development and persistence of PCS. Psychological resilience, somatic trait anxiety, embitterment, and mistrust should be taken into consideration when assessing individuals with mTBI and evaluating their risk for PCS. Targeted interventions for PCS management considering the individual personality traits could provide better care in PCS patients recovery. However, more research is needed to further understand the complex interplay between personality traits and PCS and the potential mechanisms underlying this relationship.

**Funding** None to declare.

### Declarations

**Conflict of interest** The authors declare that they have no conflict of interest.

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